

Determination of the Frequency-Dependent Impedance of a Three-phase System

Abstract—Due to the continuously varying nature of loads in the grid, it has become of great importance to determine the grid’s frequency-dependent impedance (FDI) in a fast and accurate way. In this paper, the methodology for FDI estimation is investigated, and two excitation strategies for three-phase systems are compared: (i) sequential single-phase and (ii) simultaneous three-phase perturbation. Simulations are conducted using MATLAB Simulink, covering the 0–150 kHz frequency range. Various system conditions are compared, including balanced and unbalanced systems, both with and without mutual capacitive coupling. Results show that for balanced systems with capacitive coupling, the simultaneous perturbation method loses accuracy at resonant point locations; however, under heavier loaded systems, the errors decrease. An advantage of the simultaneous method is that it injects a perturbation signal into all three phases simultaneously, thereby requiring only a third of the time compared to sequential perturbation. This could make it suitable when phases want to be compared under the same grid conditions, if the grid is relatively balanced.

Index Terms—Frequency dependent impedance, PRIS, three-phase perturbation, capacitive coupling, sequential perturbation, simultaneous perturbation.

I. INTRODUCTION

The modernisation of the power grid has introduced new challenges in power quality and stability, largely due to the widespread integration of renewable energy sources [1] and the increased use of power electronic devices in both loads and sources [2]. Inverters in systems such as photovoltaic (PV) installations exchange power with the grid at both fundamental and non-fundamental frequencies [3], emitting harmonic currents [4]. These currents interact with the grid’s frequency-dependent impedance (FDI), potentially generating harmonic voltages that can disrupt the operation of other grid-connected equipment [4] and degrade power quality. Accurate estimation of the grid’s frequency-dependent impedance (FDI) is therefore critical for system stability analysis [5]–[7], power quality assessment [8]–[10], and the reliable integration of renewable energy sources [5], [10], [11].

To obtain the FDI, the system must be excited with a suitable perturbation signal that contains frequencies throughout the entire frequency band of interest. Perturbation is categorised into non-invasive/non-intrusive and invasive/intrusive [4]. Non-invasive methods rely on natural system noise and

offer a low-cost excitation [4]. It is, however, challenging to control or repeat [9]. Invasive perturbation, on the other hand, allows precision control of excitation at a higher expense [9]. There is a wide range of invasive perturbation sources; a representative summary is shown in Figure 1 [4].

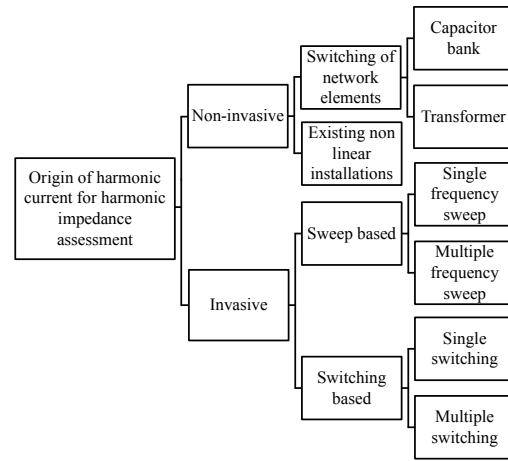


Fig. 1: Methods for system perturbation [4]

The modern grid is highly non-stationary due to factors such as switching of loads and transformer tap changers, increased number of non-linear loads, variations in the fundamental frequency and the increased installation of renewable energy sources that are interfaced to the grid using power-electronic devices [5]. The effectiveness and accuracy of an FDI measurement are affected in the case of significant load and fundamental frequency changes during measurement. This highlights the need for methods that do not require a lot of time for the FDI estimation process. Some switching loads in the grid operate at several tens of kHz [9], causing the frequency viewing range for FDI to become increasingly crucial up to at least 150 kHz [4], [9]. Furthermore, the IEC 61000-2-4:2024 also covers conducted disturbances from 0 kHz to 150 kHz [12].

Determining the FDI on a single-phase system differs from that on a three-phase system, as the latter has cross-coupling between phases. Most of the existing research has focused on FDI estimation

in single-phase systems, with little research being conducted on three-phase systems. Single-phase systems can be viewed as single-input single-output (SISO) systems, while three-phase systems exhibit multiple-input multiple-output (MIMO) behaviour due to phase coupling. Moreover, most three-phase systems operate at medium and high voltage levels [13]; which complicates the FDI measurement in such systems. The obvious challenge is the cost of components and the physical size of such a perturbation device, as illustrated in [13], where a container-sized perturbation device is employed that must be transported by truck.

This paper starts by reviewing the phase-neutral loop FDI measurement methodology for a single-phase grid. The single-phase perturbation method will be demonstrated, and the effects of a varying load on the FDI will be clearly shown. This is followed by an investigation of two methodologies for characterizing the frequency-dependent impedance in a three-phase system, namely, simultaneous and sequential perturbation. Simulations are conducted, comparing these two methods across various scenarios, including symmetrical and asymmetrical grids, as well as scenarios with coupling present in the three-phase system. This paper aims to directly compare the FDI estimates of the two mentioned methods in the phase-domain to determine if simulation perturbation could be accurate enough to be implemented in reducing measuring time for three-phase systems.

II. METHODOLOGY OF IMPEDANCE ESTIMATION TECHNIQUES

A. FDI Estimation Procedure

To determine the FDI, the discrete Fast Fourier transform is applied to the measured voltage and current, after which Ohm's law is applied to yield the FDI [8] as seen in (1)

$$Z(\omega) = \frac{\mathcal{F}\{v(t)\}}{\mathcal{F}\{i(t)\}} \quad (1)$$

where $Z(\omega)$ represents the FDI, $v(t)$ and $i(t)$ represent the measured voltage and current, respectively, in the time domain, and $\mathcal{F}$ denotes the Fourier Transform.

In practice, application of (1) directly results in errors at the fundamental frequency and its harmonics. This is due to background harmonics and the fundamental frequency being present. These need to be eliminated from the calculation to produce an accurate FDI response [4]. This is accomplished by having two measurement periods, one measuring the voltage and current at the point of observation before perturbation, and the other measuring them after perturbation [1]. This is illustrated by (2), known as the difference method [4].

$$Z(\omega) = \frac{\mathcal{F}\{v_a(t)\} - \mathcal{F}\{v_b(t)\}}{\mathcal{F}\{i_a(t)\} - \mathcal{F}\{i_b(t)\}} \quad (2)$$

The subscripts a and b indicate if the measured voltage and current were taken after or before the introduction of the perturbation signal, respectively.

B. Invasive Perturbation Signals for FDI

The choice of perturbation signal significantly influences the accuracy and reliability of FDI estimation. This paper adopts an invasive perturbation approach, in which the signals can be broadly classified into two categories: sweep-based and switching-based methods [4]. Sweep-based methods, whether employing single or multi-frequency excitation, offer high signal-to-noise ratios (SNR), enhancing the precision of system identification [14]. As noted in [7], these methods provide superior noise immunity due to the concentration of perturbation energy at a single frequency at any given time. In contrast, switching-based methods enable shorter measurement durations [14], thereby minimizing the risk of system variation during the estimation process. This makes switching-based perturbations particularly advantageous for real-time or time-sensitive FDI measurements.

III. DETERMINING FREQUENCY-DEPENDENT IMPEDANCE ON A SINGLE-PHASE SYSTEM

A. Review of Existing Broadband Perturbation

The authors in [14] investigate and compare several broadband perturbation signals, including random pulse-width modulation (PWM), discrete-interval binary sequence (DIBS), impulse signals, and pseudorandom binary sequence (PRBS), before proposing the PRIS signal, which is derived from the PRBS. The PRIS source features a tunable RLC filter that can decrease the overall power loss, as well as the current ratings of the switching components needed for the PRIS source, making it suitable for high-power, high-voltage system identification applications [14]. In an experiment, the authors showed that the PRIS source achieved 14 times lower average power loss than a PRBS source [14]. Based on these advantages, the PRIS is selected as the perturbation signal in this paper.

B. Simulation Model: Single-Phase System

The system under consideration comprises a source with its associated series impedance, with parameter values selected to represent a strong grid, followed by a cable and a load representative of a typical domestic customer. The customer load is modelled as a parallel combination of RC and RL branches, as described in [15] and illustrated in Figure 2. The specific component values used in the simulation are provided in Table I.

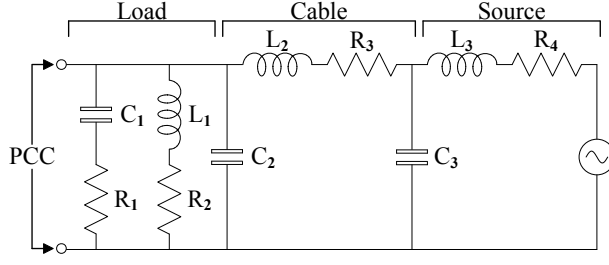


Fig. 2: Single-phase system model

TABLE I: System model parameters

Component	Value
R_1, R_2, R_3, R_4	$0.5\Omega, 130\Omega, 0.7\Omega, 0.173\Omega,$
L_1, L_2, L_3	$95\text{mH}, 0.981\text{mH}, 0.4822\text{mH}$
C_1, C_2, C_3, V_{AC}	$5\mu\text{F}, 1\mu\text{F}, 1\mu\text{F}, 220V_{rms}$

C. Simulation Arrangement: Single-Phase Network

The objective of this section is to evaluate the impact of varying load conditions on the FDI. The FDI is determined from the perspective of the Point of Common Coupling (PCC); hence, the PRIS perturbation source is configured to inject signals at the PCC towards the source. The Simulink model is configured to employ a fixed-step solver with automatic solver selection and a fixed time step of $1 \mu\text{s}$. The *powergui* block is set to discrete simulation mode with a sample time of $1 \mu\text{s}$. Load conditions are varied to represent scenarios with 0, 1, 10, and 100 parallel-connected customers. To assess the accuracy of the FDI estimation, the impedance is calculated analytically to use as a reference, this is shown in equation (3).

$$Z = \left(\frac{1}{\alpha + L_2 + R_3} + \frac{1}{C_2} + \frac{1}{L_1 + R_2} + \frac{1}{C_1 + R_1} \right)^{-1} \quad (3)$$

$$\alpha = \frac{(L_3 + R_4) \cdot C_3}{C_3 + L_3 + R_4}$$

D. Simulation Results: Single-Phase Network

Figure 3 presents the FDI results across the full frequency range up to 150 kHz, along with a detailed view of the resonant points near 10 kHz. In the no-load scenario, three distinct resonance points are observed. A parallel resonance P_1 at approximately 3.9 kHz caused by the parallel combination of $C_2 || (L_2 + L_3)$. A series resonance S_1 at around 8.8 kHz caused by the series combination of $(L_2 || L_3) + C_3$. Lastly, a second parallel resonance P_2 at roughly 9.4 kHz caused by the parallel combination $(C_2 + C_3) || L_3$.

When comparing the no-load case to the scenario with a single customer, it is observed that P_1 is lightly damped and shifts slightly to a lower frequency. Meanwhile, S_1 and P_2 remain at approximately the same frequencies but exhibit increased damping. This trend intensifies as more customer loads are added; all resonance points

become progressively more damped. At 100 customers, both S_1 and P_2 are so heavily damped that they are no longer distinguishable in the response. These findings provide valuable insights for system integrators seeking to evaluate the influence of aggregated customer loads on grid stability and power quality.

These results further indicate how a varying load can significantly alter the FDI of the system, underscoring the need for a rapid in-field FDI test.

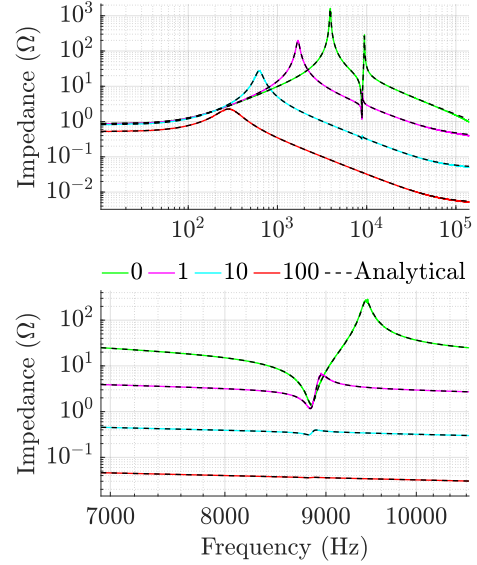


Fig. 3: FDI for single-phase network with varying load

IV. GRID FREQUENCY-DEPENDENT IMPEDANCE ON A THREE-PHASE SYSTEM

A. Literature Review of Perturbation Methods

In a three-phase system, three popular perturbation methods have emerged on how to perturb the system. These are, namely, single-phase, line-to-line, and three-phase perturbation [7]. Figure 4 illustrates these three methods, focusing on a four-wire system, where arrows indicate the perturbation point and the return path for the current.

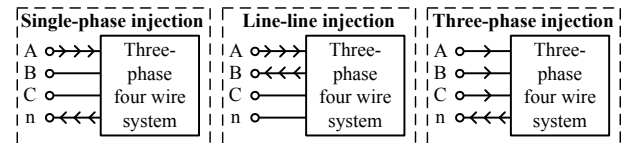


Fig. 4: Perturbation methods for three-phase systems.

The research in [16] focuses specifically on comparing the three perturbation methods when calculating the FDI for overhead transmission lines using fixed-frequency sinusoidal perturbation. The measurements reported by

[16] are conducted offline, but are still valuable as they provide insight into the impedance obtained from each perturbation method. It is shown that the full FDI matrix for a three-phase system is represented as in (4).

$$\mathbf{Z}_f = \begin{bmatrix} \frac{U_{aa}}{I_a} & \frac{U_{ab}}{I_a} & \frac{U_{ac}}{I_a} \\ \frac{U_{ba}}{I_b} & \frac{U_{bb}}{I_b} & \frac{U_{bc}}{I_b} \\ \frac{U_{ca}}{I_c} & \frac{U_{cb}}{I_c} & \frac{U_{cc}}{I_c} \end{bmatrix} = \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix} \quad (4)$$

The voltage U_{xy} represents the voltage measured at phase y when phase x is perturbed. The current I_x represents the current measured flowing into phase x when that phase is perturbed. In their case, the single-phase sequential method is the only method that can calculate the full impedance matrix shown in (4) [16]. This is due to only one phase being perturbed at a time, making it possible to measure the effect of perturbation on one phase on the other phases.

B. Perturbation Methodology for Three-Phase Systems

Since both line-line and three-phase perturbations excite more than one phase at a time, the effect of one phase's perturbation on another phase is not possible to isolate. Due to this, in [16], only the single-phase perturbation method delivers both positive and zero sequence impedance results. In contrast, the line-line and three-phase perturbation methods deliver only positive sequence results. However, if there is only an interest in the positive sequence impedance or if the system is relatively well-balanced, the three-phase perturbation method can be applied, reducing the measurement time needed to one-third of that required for the single-phase method.

C. Line Model with Capacitive Coupling for Three-phase Systems

The single-phase model in Figure 2 is expanded into three phases, with three voltage sources 120° out of phase relative to each other and the neutral connected. Mutual capacitive coupling is introduced to the transmission line to more accurately reflect real-world behaviour. The authors in [17] discuss a three-phase π circuit, while [18] discusses a ladder network. The authors in [18] mention that to obtain symmetry, the lateral capacitance can be divided into two halves on both sides of the line model. Taking the knowledge of both [17] and [18], Figure 5 shows the model that includes the capacitive coupling for a transmission line model.

D. Coupling Capacitor size selection

Several sources were consulted to determine the relative component sizes of mutual capacitors on three-phase transmission lines. The sizes of the capacitors are shown in Table II.

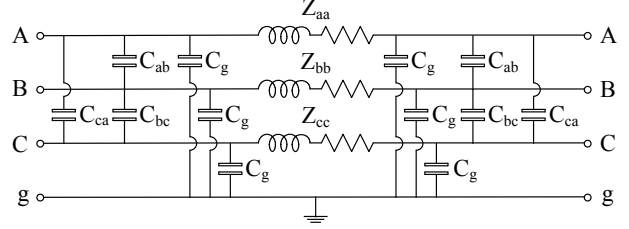


Fig. 5: Three-phase line model with mutual capacitive coupling between phases

TABLE II: Capacitor information from sources

Source	V_{rated}	C_{self}/km	C_{mut}/km	Ratio %
[19]	345 kV	27.62 nF	7.75 nF	28.1
[20]	500 kV	10.6 nF	2.24 nF	21.1
[21]	—	11.08 nF	1.18 nF	10.6

It was decided to choose the mutual conductance to be conservative 10 % of the value of the self-capacitance. The mutual capacitance of the line model was therefore chosen to be $0.2 \mu F$ and split in half on both sides [18] therefore, the values of the mutual capacitors C_{ab} , C_{bc} , and C_{ca} are $0.1 \mu F$.

E. Simulation Arrangement: Three-Phase System

In the MATLAB Simulink simulation environment, a total simulation time of 10 seconds was used, during which the three phases, A , B , and C , were individually and then simultaneously perturbed using the PRIS. This was done to evaluate both single-phase and three-phase perturbation methods, as illustrated in Figure 6.

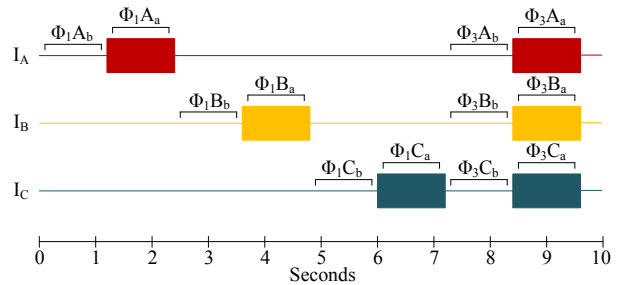


Fig. 6: Simulation perturbation sequence and measurement intervals

The three separate line graphs shows the response of the phase's current during the simulation. When perturbation is applied the measured current on the phase increases drastically due to the PRIS device injecting current into that phase. ϕ_1 and ϕ_3 indicate the windows that are used for the voltage and current measurements for single-phase and three-phase calculations, respectively for phase A , B and C . A subscript b and a denote the measurements before perturbation and during perturbation, respectively. Each

phase operates as normal without being perturbed for 1.2 seconds, after which it is perturbed by the PRIS for 1.2 seconds. The measurements are done for 1 second inside the 1.2 seconds window, as shown in Figure 6. This ensures that no transient noise affects the measurements. Equation (2) is then applied to calculate the FDI. The load was varied in the same manner as used in the single-phase system experiment. Simulink's *Impedance Measurement* block is implemented to calculate the correct FDI to use as the reference, as analytical calculations become more complex with the added capacitive coupling.

F. Results: Balanced System Without Mutual Coupling

Due to the equal loads on all three phases, the FDI of each phase is identical. Using the *A* phase, the simulation yielded the same graph as displayed in Figure 3 with the sequential and simultaneous perturbation following the analytical calculation. This indicates that, in the ideal case with balanced load and no coupling present, simultaneous perturbation yields the same result as sequential perturbation.

G. Results: Unbalanced System Without Mutual Coupling

In this simulation, the load was varied with phases to create imbalance. Phase *A* was connected to 1 customer, *B* to 10 customers, and *C* to 100 customers. Once again, the sequential and simultaneous methods both yielded the same response and followed the reference as in Figure 3.

H. Results: Balanced System with Mutual Coupling

In this simulation, the coupling capacitors in Figure 5 are included. The load is varied in the following order: no load, 1 customer, 10 customers, 100 customers. Figure 7 shows the sequential perturbation results only looking at phase *A* due to the phases being balanced. It is

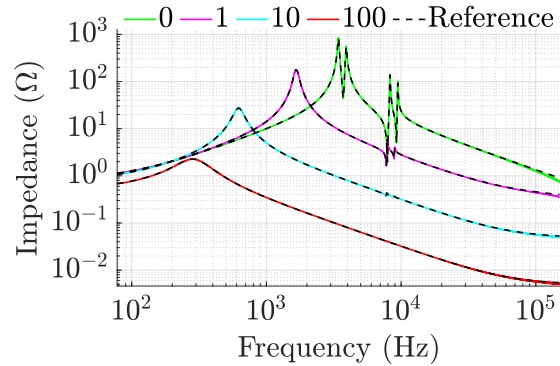


Fig. 7: FDI of balanced and mutually coupled system using sequential perturbation method

noted that the sequential perturbation method closely

follows the reference throughout the entire 150kHz frequency range. Two new series and parallel resonant points appear, caused by the coupling capacitors, and are very evident with the no-load and 1 customer results. Figure 8 shows the simultaneous perturbation results.

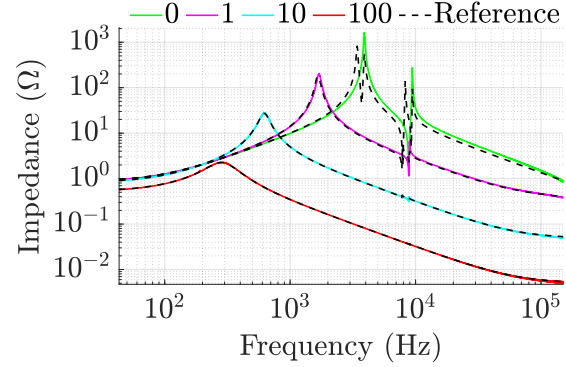


Fig. 8: FDI of balanced and mutually coupled system using simultaneous perturbation method

With the simultaneous perturbation method, the estimated FDI closely matches the analytical reference in terms of the general shape. It, however, fails to capture the effect of coupling impedance between the phases. However, this limitation is only significant under minimal loading conditions. As the number of customers increases, the resonant effects induced by mutual coupling are increasingly damped, and the simultaneous perturbation results once again align well with the reference in the cases of 10 and 100 customers.

I. Results: Unbalanced System with Mutual Coupling

In this simulation, the three phases are assigned different load levels to emulate an unbalanced network with mutual coupling. Specifically, phase *A* is loaded with 1 customer, phase *B* with 10 customers, and phase *C* with 100 customers. The corresponding results are shown in Figure 9. In this case, the FDI of

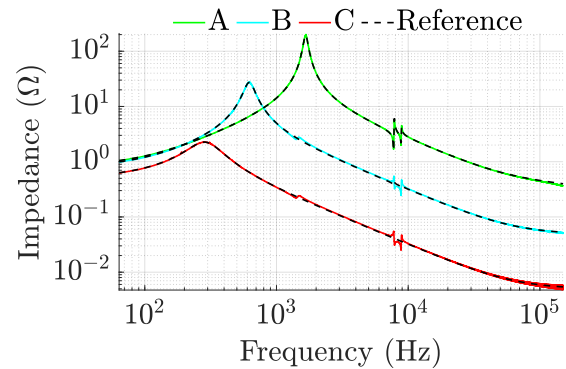


Fig. 9: FDI of unbalanced and mutually coupled system using simultaneous perturbation method

phase A is accurately estimated; however, the FDI estimates for phases B and C are distorted by resonance effects originating from phase A in the 7 kHz–10 kHz range. This illustrates that, in unbalanced systems, the simultaneous perturbation method is susceptible to perturbation leakage across phases. The imbalance causes uneven current sharing via mutual coupling, leading some phases to exhibit stronger induced responses than others, thereby compromising the accuracy of FDI estimation.

V. CONCLUSION

This paper investigates frequency-dependent impedance (FDI) estimation in single-phase and three-phase systems, focusing on the comparison between sequential and simultaneous perturbation methods. Under both balanced and unbalanced conditions without capacitive coupling, the two methods produced identical FDI results. However, in the presence of capacitive coupling between phases, the simultaneous method became inaccurate under light loading. As the load increased, the impact of the coupling capacitors was damped, and the simultaneous method aligned with the sequential results. In unbalanced systems with mutual coupling, the phase with the smallest load yielded accurate FDI, while the other phases exhibited errors near the resonant frequencies of the lightly loaded phase. Despite these limitations, simultaneous perturbation remains viable in relatively balanced systems or when other network elements sufficiently damp mutual coupling effects. Additionally, simultaneous-based perturbation ensures identical grid conditions across phases, enhancing result comparability. This study focuses solely on capacitive coupling, and inductive coupling is not included. Future work will explore expanded models that incorporate mutual inductance to better reflect real-world network behaviour.

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